

Library Management System – SQL Project Report

1. Overview

A complete SQL-based system built to manage branches, employees, members, books, issue and return operations. Core focus areas include database design, CRUD operations, analytical queries, CTAS usage, and procedures for automation.

2. Database Schema

Key Tables:

- branch
- employees
- members
- books
- issued_status
- return_status

3. Core Functional Features

- Insert, update, delete operations for managing library records.
- Issue and return handling with constraints.
- Book availability tracking.
- Branch-level operational insights.

4. Analytical Queries

Insights derived:

- Members with multiple issues.
- Revenue by book category.
- Books currently not returned.
- Issue frequency per title.

5. CTAS

Created datasets for:

- Book issue frequency table for reporting.
- Branch-level operational summary.

6. Stored Procedures

Two major procedures:

- add_return_records: Automates updating return records and book availability.
- issue_book: Validates availability and processes issuing workflow.

7. Reports & Insights

Branch performance summary includes issues, returns, and revenue metrics. Useful for operational decisions.

8. Conclusion

The system demonstrates:

- Real-world relational design
- Hands-on SQL proficiency
- Reporting and automation capability

Suitable for portfolio demonstration and interview assessment.